

ECON 139: Labor Economics

Problem Set 2

Note on Grading: Homework is graded using check +, check or check -. If you answer all questions and generally show you have put thought and effort into the problem set, you will get a check + (1/1). If your answers are sometimes incomplete, or show a clear lack of effort in parts, you will receive a check (0.7/1). If your answers are often incomplete and show little effort overall, you will get a check - (0/1). The total number of points for each problem set in the course is 1 point and each problem set is worth 5 percent of the grade. If you turn this in within one day late, the highest score you can get is a check (0.7). No credit is given for assignments that are turned in more than one day late. This problem set due is Wednesday November 19th.

Note on AI Use: As discussed in the syllabus, you can use AI tools to help with this problem set. For the first part, however, I highly recommend you complete this on your own without any AI assistance. This type of problem may very well appear on a midterm or final. If you cannot solve it yourself, you will likely struggle on the exam.

For the empirical part, I recommend using AI tools. Not everything in the empirical question comes directly from the lecture. You will need to figure out some parts on your own (as you will often need to do in the future!). AI tools will be useful for this.

1 Part 1: Theoretical Exercise

1. Imagine there are two types of workers: College Graduates (CG) and Non-College Graduates (NCG). College Graduates generally earn more than Non-College Graduates. Throughout the 1980s and 1990s, the fraction of individuals with a college degree increased significantly. Assuming nothing else changed, how would expect the relative wages of CG and NCG to evolve over this period? Explain your reasoning using supply and demand diagrams (you can assume perfectly competitive labor markets).
2. Imagine a firm needs to fill three tasks to produce one unit of output. For example, the tasks could be: (1) assembling the product, (2) managing inventory and (3) designing the product/overseeing production processes. The firm can hire two types of workers: High-Skill (HS) workers who can perform all three tasks, and Low-Skill (LS) workers who can perform tasks (1) and (2). Below we present a table that shows the cost of hiring each type of worker per task. The cost is different for different tasks because it may take more time for a worker to perform a certain task relative to another task.

Worker Type	Task 1 Cost	Task 2 Cost	Task 3 Cost
High-Skill (HS)	\$20	\$40	\$40
Low-Skill (LS)	\$10	\$20	—

What is the total cost of producing one unit of output assuming the firm wants to minimize costs?

3. Imagine now that there is a secular increase in the supply of High-Skill workers. This decreases the cost of hiring High-Skill workers to: \$15 for Task 1, \$30 for Task 2 and \$30 for Task 3. Do you think the firm will change how much it is producing? You can assume that the more output the firm produces, the less it can sell each unit of output for (i.e. the demand curve for the firm's output is downward sloping). Explain your reasoning.
4. How does the change in the supply of High-Skill workers affect the demand for Low-Skill workers? Explain your reasoning.
5. Now imagine there is a new machine that the firm can rent to perform Task 1. Imagine the cost of renting this machine is \$9 per unit of output. How does this affect the firm's demand for High-Skill and Low-Skill workers? Explain your reasoning. The new cost structure for the firm is below:

Input Type	Task 1 Cost	Task 2 Cost	Task 3 Cost
High-Skill (HS)	\$20	\$40	\$40
Low-Skill (LS)	\$10	\$20	—
Machine	\$9	—	—

6. Now imagine the cost of the machine per unit is very low (because the machine can assemble products so quickly). For example, imagine the cost of renting the machine is only \$2 per unit of output. How does this affect the firm's demand for High-Skill and Low-Skill workers relative to the prior question? Explain your reasoning.

2 Part 2: Empirical Exercise

In this empirical exercise, we are going to return to O*NET data to explore tasks performed by occupations. You will need to download a data file named “Tasks” (the relevant file you should download depends on whether you complete the empirical exercise with Excel (xlsx), Stata (dta), R (.csv) or Python (.csv)).

To complete this exercise, you will need to turn in (1) a writeup of the results, which includes the figures you generate, and any answers to questions below, and (2) the code you used to generate your figure (if applicable). If you use Excel, upload the completed Excel file with your writeup.

Note: The structure of the file is similar to the file we looked at in class. But instead of “Work Activities”, this file contains information on “Tasks”. If you are confused about the structure of the data, please refer back to the empirical exercise we did in class on work activities (Week 6 Lecture on O*NET Data).

- Go to ONET Online and search for any occupation that you are interested in. This could be the occupation that you will be presenting on for your occupation presentation or an occupation you are curious about. Once you find the occupation, retrieve the O*NET SOC Code.
- Load in the “Tasks” data and restrict it to the occupation that you are interested in.
- Create a figure that displays the top 5 tasks performed by this occupation. This should be a figure (not a table), but should allow the reader to be able to (1) easily see which tasks are the most important and (2) see how important each of the top 5 tasks are relative to one another. Make sure to label your axes and provide a title for your figure.
- For each of the top 5 tasks, categorize this task as routine or nonroutine and cognitive or manual. For each one describe whether you think it could be automated by AI.
- Retrieve the Task ID associated with the most important task for your occupation. Then, reload the original dataset (which contains information from all occupations) and restrict to this Task ID.
- Create a histogram that shows the distribution of importance of this task across all occupations. Make sure to label your axes and provide a title for your figure. Add a vertical line that shows the importance of this task for your chosen occupation.